The Kuratowski–Bekenstein–Hawking Translation: Discrete Foundations, Continuum Conjectures, and Negative Results

A Machine-Verified Discrete Theory with a
Precise Research Program for Continuum Reconstruction

Version 2: corrected and restructured.
Replaces v1 (same DOI).

J. P. Silva Alvarado

DOI: 10.5281/zenodo.19198183

2026

Abstract

We present the *Kuratowski Calculus* — a discrete physics on Poisson-sprinkled causal sets — together with its conjectural translation into continuum quantum field theory via the Bekenstein–Hawking entropy bound.

The discrete theory is machine-verified: all combinatorial foundations are proved in Lean 4 (zero sorrys on the main path) or verified by exhaustive computation. The theory predicts hadron and lepton mass ratios to $< 2\%$ accuracy with zero free parameters.

The continuum translation is **incomplete**. We identify three precise open problems that must be solved before a rigorous continuum QFT can be constructed: (A) defining the correct ambient state space for topological excitations, (B) proving a spectral gap for the defect/rewrite Hamiltonian, and (C) reconstructing a continuum relativistic QFT from the discrete data.

We also report a significant **negative result**: the extension graph Laplacian, previously proposed as the mass-gap operator, has spectral gap $\lambda_1 \sim \pi^2/n^2 \to 0$ universally, regardless of Hasse degree. This rules out the extension graph as the source of a uniform mass gap and necessitates a different operator.

Every claim in this document is tagged with its provenance: [Lean] (Lean 4 verified), [Exhaustive] (exhaustive computation), [Computational] (computational observation), [Model] (model prediction), or [Conjecture] (conjecture).

Contents

1 The Problem **3**

2 The Discrete Theory **3**

3 The Continuum Axiom **4**

4 From the Axiom to Bounded Degree **4**

5 The Translation Dictionary **5**

6 Combinatorial Foundations (Discrete — Established) 6
6.1 The A/F/R decomposition . . . 6
6.2 Extension Graph connectivity . . . 6
6.3 Hasse diagram properties . . . 6
6.4 The Kirchhoff mass observable . . . 6
6.5 The Weak Transfer Bound . . . 7
6.6 Balance Preservation . . . 7

7 The Kuratowski Vacuum 7
7.1 The continuum Hilbert space (W0) . . . 8

8 Field Operators 10

9 Wightman Axioms 10
9.1 W1: Poincaré covariance . . . 10
9.2 W2: Spectral condition . . . 10
9.3 W3: Unique vacuum . . . 10
9.4 W4: Locality . . . 11
9.5 W5: Completeness . . . 11

10 Mass Gap 11

11 Gauge Group 11

12 Non-Triviality 12

13 The Bekenstein–Hawking Quarter 12

14 Fine Structure Constant 13

15 Lepton Mass Spectrum 14

16 Hadron Spectrum 14

17 Chiral Phase Conservation and Lepton Precision Masses 15
17.1 The chiral $K_{2,2}$ . . . 15
17.2 Bridge types . . . 15
17.3 Phase conservation equations . . . 16
17.4 Solutions . . . 17

18 Identification with Yang–Mills 17
18.1 Structure constants . . . 18
18.2 Hadron spectrum . . . 18
18.3 Vacuum polarisation and coupling evolution . . . 18
18.4 Perturbative identification . . . 19
18.5 Discrete asymptotic freedom . . . 19

19 Main Theorem 20

20 Rigour Audit **21**
20.1 What is machine-verified . . . 21
20.2 What depends on BH . . . 21
20.3 What is the single axiom . . . 22

21 Predictions Beyond the Prize **22**

1 Scope and Status

The Clay Mathematics Institute asks [1]:

> **Yang–Mills Existence and Mass Gap.** *Prove that for any compact simple gauge group G, a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ and has a mass gap $\Delta > 0$. Existence includes establishing axiomatic properties at least as strong as those cited in* [2] *and* [3].

This paper does not solve the Clay problem. It presents a candidate discrete theory with machine-verified combinatorial foundations, a heuristic translation dictionary to continuum physics, and a precise research program for future continuum reconstruction.

The document has three layers:

1. **Discrete foundations** (completed, machine-verified): a finite combinatorial theory on Poisson-sprinkled causal sets, proved in Lean 4 or verified by exhaustive computation.
2. **Translation dictionary** (heuristic): a correspondence table between discrete objects and continuum physics. This is a guide for future work, not a theorem.
3. **Continuum conjectures** (open): three precisely stated problems whose resolution would yield a rigorous continuum QFT.

An earlier version of this document (v1, same DOI) overclaimed the continuum results as proved. This version corrects those claims.

2 The Discrete Theory

The *Kuratowski Calculus* [12] is a discrete physics built on Poisson-sprinkled causal sets [4]. We summarise its axiom and key objects; full definitions and proofs are in [12], with machine verification in Lean 4 (accompanying files).

Definition 2.1 (Causal set and Hasse diagram)**.** *A* causal set *is a finite partial order $(P, \preceq)$ obtained by Poisson sprinkling N points into a bounded region of Minkowski spacetime $\mathbb{R}^{3,1}$, with $x \preceq y$ iff $y - x$ is future-causal. The* Hasse diagram *$H(P)$ is the directed graph of covering relations: $x \to y$ iff $x \prec y$ and no z satisfies $x \prec z \prec y$.*

Definition 2.2 (Axiom 0 — Volumetric Limit)**.** *Every vertex v of the Hasse diagram $H(P)$ has bounded degree:*

$$\deg_H(v) \;\leq\; \Delta_{\max},$$

where $\Delta_{\max}$ is a universal constant depending only on the spacetime dimension. For $d = 4$: $\Delta_{\max} = 15$ (measured in the kvac engine at $N = 10^7$, [12]).

This is the single axiom of the discrete theory. All physics — the gauge group, confinement, the mass gap, the particle spectrum — follows from it.

Remark 2.3. *Two immediate consequences of Axiom 0:*

1. *The Hasse diagram is* ***triangle-free****: if $u \lessdot w \lessdot v$ are covering relations, then $u \lessdot v$ is not a covering (it is redundant). Lean-verified:* `covers_triangle_free` *in* `HasseDAG.lean`*.*

2. *The Hasse diagram has bounded local complexity: every vertex interacts with at most $\Delta_{\max}$ neighbours. The vacuum (flat spacetime) is* ***locally series-parallel****: zero non-planar components. Non-planarity is a topological* ***excitation*** *above the vacuum.*

3 The Continuum Axiom

Axiom 1 (Holographic Volumetric Limit)**.** *For every bounded causal diamond $\Omega \subset \mathbb{R}^{3,1}$ with boundary area $A(\partial\Omega)$, the physical Hilbert space $\mathscr{H}_\Omega$ satisfies:*

$$\dim(\mathscr{H}_\Omega) \;\leq\; \exp\left(\frac{A(\partial\Omega)}{4\,\ell_{\mathrm{P}}^2}\right), \tag{1}$$

where $\ell_{\mathrm{P}} = \sqrt{G\hbar/c^3}$ is the Planck length.

This is the Bekenstein–Hawking entropy bound [5, 6] restated as a bound on Hilbert space dimension. It is the continuum translation of the discrete Axiom 0 (Definition 2.2):

Discrete (Axiom 0)	**Continuum (Axiom 1)**
Every Hasse node has $\deg(v) \leq \Delta_{\max}$	Every causal diamond has $\dim(\mathscr{H}) \leq e^{A/4\ell_{\mathrm{P}}^2}$
Bounded local information	Bounded regional information
$\Delta_{\max} = 15$ (measured)	$A/4\ell_{\mathrm{P}}^2$ (Bekenstein–Hawking)
Consequence: triangle-free Hasse	Consequence: finite-dimensional

Remark 3.1 (Relationship between axioms)**.** *Axiom 1 and the discrete Axiom 0 (Definition 2.2) are conjectured to be the same physical statement at different resolutions. The discrete bound $\deg(v) \leq 15$ counts neighbours per Planck cell; the continuum bound counts bits per boundary area.*

Caveat: *Theorem 4.1 derives bounded Hasse degree from Axiom 1, but requires two additional hypotheses: a sublinear local area growth law $A_{\mathrm{local}}(v) \leq C_{\mathrm{loc}}\, k^\beta\, \ell_{\mathrm{P}}^2$ and the assumption that each neighbour contributes one independently switchable binary degree of freedom. These are physical modelling postulates, not consequences of BH alone. The "single axiom" framing of v1 was an overclaim.*

4 From the Axiom to Bounded Degree

The discrete theory uses bounded Hasse degree as its foundation. Here we derive it from Axiom 1.

Theorem 4.1 (Bounded degree from BH + local area growth)**.** *Let $\mathcal{C}_\rho$ be a Poisson-sprinkled causal set in $\mathbb{R}^{3,1}$ at density ρ, and let $H(\mathcal{C}_\rho)$ be its Hasse diagram. Assume Axiom 1 and assume there exist constants $C_{\mathrm{loc}} > 0$ and $\beta \in (0,1)$ such that for every vertex v with $k = \deg_H(v)$,*

$$A_{\mathrm{local}}(v) \;\le\; C_{\mathrm{loc}}\, k^\beta\, \ell_{\mathrm{P}}^2,$$

where $A_{\mathrm{local}}(v)$ is the boundary area of the smallest causal diamond containing v and all its Hasse neighbours, and each neighbour contributes one independently switchable binary degree of freedom. Then

$$\Delta_{\max} \;\le\; \left(\frac{C_{\mathrm{loc}}}{4\ln 2}\right)^{1/(1-\beta)}.$$

Proof. Fix a vertex v with $k = \deg_H(v)$. The k neighbours contribute 2^k binary configurations. By Axiom 1:

$$2^k \;\le\; \exp\!\left(\frac{A_{\mathrm{local}}(v)}{4\ell_{\mathrm{P}}^2}\right) \qquad\Longrightarrow\qquad k \;\le\; \frac{A_{\mathrm{local}}(v)}{4\ell_{\mathrm{P}}^2 \ln 2}.$$

Substituting the local growth hypothesis $A_{\mathrm{local}}(v) \le C_{\mathrm{loc}}\, k^\beta\, \ell_{\mathrm{P}}^2$:

$$k \;\le\; \frac{C_{\mathrm{loc}}}{4\ln 2}\, k^\beta \qquad\Longrightarrow\qquad k^{1-\beta} \;\le\; \frac{C_{\mathrm{loc}}}{4\ln 2} \qquad\Longrightarrow\qquad k \;\le\; \left(\frac{C_{\mathrm{loc}}}{4\ln 2}\right)^{1/(1-\beta)}.$$

This bound is uniform in v, so $\Delta_{\max}$ is finite. In $d = 4$, taking $\beta = 1/2$ gives $\Delta_{\max} \le (C_{\mathrm{loc}}/(4\ln 2))^2$. The measured value $\Delta_{\max} = 15$ [12] gives $C_{\mathrm{loc}} = 4\ln 2 \cdot 15^{1/2} \approx 10.7$. □

Remark 4.2. *The value $\Delta_{\max} = 15$ was measured independently in the discrete vacuum (`MAX_HASSE_DEGREE` in the kvac engine [12]). Inserting $\beta = 1/2$ and $\Delta_{\max} = 15$ into the bound yields $C_{\mathrm{loc}} \approx 10.7$, a consistency check between the local area-growth scaling and the measured degree cap.*

5 Translation Dictionary (Heuristic)

The discrete Kuratowski Calculus is a completed combinatorial theory. The following table lists **proposed** correspondences between discrete objects and continuum physics. These identifications are heuristic guides for future work, not proved equivalences. [Model]

#	Discrete	Continuum	Bridge
0	$\deg(v) \le 15$	$S \le A/4\ell_{\mathrm{P}}^2$	BH axiom
1	Poset P on n elements	Causal diamond $\Omega \subset \mathbb{R}^{3,1}$	Poisson sprinkling
2	$\ell^2(L(P))$, dim $= k$	$\mathscr{H}_\Omega$, dim $\le e^{A/4\ell_{\mathrm{P}}^2}$	BH bounds dim
3	Laplacian of $\mathcal{E}(P)$	Hamiltonian H_Ω	Graph $\to$ operator
4	$K_{3,3}$ subdivision	Non-planar defect, area $\ge A_{\min}$	BH gives area cost
5	$\tau(K_{3,3}) = 81$	Mass gap $\Delta > 0$	Kirchhoff
6	$Q_{\mathrm{topo}} = 1/4$	BH coefficient 1/4	§13

#	Discrete	Continuum	Bridge
7	$\alpha_0 = 1/(32\pi)$	Bare coupling at Planck scale	$Q_{\text{topo}}/(8\pi)$
8	Vacuum screening $32\pi/137$	Vacuum polarisation	RG flow
9	Genus ladder $g = 0, 1, 2$	Topological handles	BH area cost
10	Rewrite phase (de-priming)	Vacuum fluctuations	Virtual pairs
11	Series-parallel vacuum	Flat spacetime ($R_{\mu\nu\rho\sigma} = 0$)	Planarity = flatness
12	$A/F/R$ partition	UV / IR / confined modes	§6.1
13	Balance $S(a, b) = c/k$	Wightman 2-point function	§8
14	Flexible involution	Spacelike commutation (W3)	§9.4
15	$\mathcal{E}(P)$ connected	Unique vacuum (W2)	§9.3
16	$K_{2,n}$ backbone + genus	Lepton spectrum	§15
17	Phase Conservation	Hadron spectrum	§16

6 Combinatorial Foundations (Discrete — Established)

All results in this section are proved in Lean 4 with zero sorrys, or are computationally exhaustive. We state them without proof; the proofs are in [12].

6.1 The A/F/R decomposition

Theorem 6.1 (A/F/R partition, [12])**.** *For any finite poset P and incomparable pair $a \parallel b$, the set $L(P)$ of linear extensions partitions into six cells:*

$$L(P) = A^+ \sqcup F^+ \sqcup R^+ \sqcup A^- \sqcup F^- \sqcup R^-,$$

where $A^\pm$ are adjacent *($|buffer| = 0$), $F^\pm$ are* flexible *(buffer elements all incomparable to a and b), and $R^\pm$ are* rigid *(some buffer element comparable to a or b). The $\pm$ denotes whether a precedes b or vice versa.*

Theorem 6.2 (Involution identities, [12])**.** $|A^+| = |A^-|$ *and* $|F^+| = |F^-|$.

Corollary 6.3 (Skewness equation)**.** $c(a, b) - c(b, a) = |R^+| - |R^-|$. *All balance information is carried by the rigid linexts.*

6.2 Extension Graph connectivity

Theorem 6.4 ([12, ExtGraph.lean])**.** *The Extension Graph $\mathcal{E}(P)$ (vertices = $L(P)$, edges = adjacent bottleneck swaps) is connected for every finite poset P.*

6.3 Hasse diagram properties

Theorem 6.5 (Triangle-freeness, [12, HasseDAG.lean])**.** *The undirected graph underlying any Hasse diagram is triangle-free.*

Theorem 6.6 (Kuratowski obstruction)**.** *Every non-planar graph contains a subdivision of $K_{3,3}$ or K_5 (Kuratowski 1930 [7]). Both obstructions can occur in triangle-free Hasse diagrams.*

6.4 The Kirchhoff mass observable

Theorem 6.7 ($\tau(K_{3,3}) = 81$, $\tau(K_5) = 125$, [12, K33.lean])**.** *The reduced Laplacian determinants are $\tau(K_{3,3}) = 81$ and $\tau(K_5) = 125$. Both are strictly positive. The $K_{3,3}$ value is verified by* `native_decide` *in Lean 4.*

Corollary 6.8 (Universal non-planar mass bound)**.** *For any connected non-planar graph G:*

$$\tau(G) \;\geq\; \min(\tau(K_{3,3}),\, \tau(K_5)) \;=\; \min(81, 125) \;=\; 81 \;>\; 0.$$

Proof. By Kuratowski, G contains a subdivision H of $K_{3,3}$ or K_5.

First, subdivision does not decrease the tree number: if H' is obtained from a connected graph H_0 by subdividing one edge e, deletion–contraction on one of the two new edges gives $\tau(H') = \tau(H_0)+\tau(H_0-e) \geq \tau(H_0)$. Iterating over subdivision steps yields $\tau(H) \geq \tau(K_{3,3})$ or $\tau(H) \geq \tau(K_5)$.

Second, H is a subgraph of G. For every edge f, deletion–contraction gives $\tau(G) = \tau(G-f) + \tau(G/f) \geq \tau(G-f)$. Iterating edge deletions from G down to H yields $\tau(G) \geq \tau(H)$.

Therefore $\tau(G) \geq \tau(H) \geq \min(\tau(K_{3,3}), \tau(K_5)) = \min(81, 125) = 81$. □

Remark 6.9 (K_5 exclusion is unnecessary)**.** *The mass gap $\tau \geq 81 > 0$ holds for* both *Kuratowski obstructions. Prior versions of this argument claimed that K_5 subdivisions do not occur in causal Hasse diagrams. This claim is false: explicit counterexamples exist (e.g., 5 antichain elements with 10 common upper covers, or the Petersen graph oriented as a DAG). The corrected theorem does not need K_5 exclusion: the mass gap is universal.*

6.5 The Weak Transfer Bound

Theorem 6.10 ([12])**.** *For any finite poset P and incomparable pair $a \parallel b$: if the Weak Transfer Bound holds —*

$$S_{\text{sym}} + 2\min(R^+, R^-) \geq \max(R^+, R^-)$$

— then $1/3 \leq S(a,b) \leq 2/3$.

Conversely, $1/3 \leq S(a,b) \leq 2/3$ implies the Weak Transfer Bound. The two statements are algebraically equivalent.

Remark 6.11. *The Weak Transfer Bound has been verified on* 423,506 *unique prime posets at $n = 4, \ldots, 8$ with zero violations.*

6.6 Balance Preservation

Theorem 6.12 ([12, Clustering.lean])**.** *Let P be a finite poset, $x \parallel y$ in P, and $z \notin P$ with $z \parallel x$ and $z \parallel y$. Then $c_{P\cup\{z\}}(x,y)/k_{P\cup\{z\}} = c_P(x,y)/k_P$.*

Elements that are incomparable to both x and y do not change the balance. This is the key to convergence.

7 The Kuratowski Vacuum

Definition 7.1 (Causal set)**.** *Fix a bounded causal diamond* $\mathbf{A} \subset \mathbb{R}^{3,1}$*. A* Kuratowski vacuum at density ρ *is a Poisson point process* $\mathcal{C}_\rho$ *of intensity* ρ *on* $\mathbf{A}$*, with the causal partial order* $\preceq$ *from the Minkowski metric.*

Definition 7.2 (Hilbert space)**.** $\mathscr{H}_\rho = \ell^2(L(P_\rho))$*, with orthonormal basis* $\{|\sigma\rangle : \sigma \in L(P_\rho)\}$*.*

Definition 7.3 (Hamiltonian)**.** $H_\rho =$ *graph Laplacian of* $\mathcal{E}(P_\rho)$*:*

$$H_\rho|\sigma\rangle = \deg(\sigma)|\sigma\rangle - \sum_{\tau\sim\sigma} |\tau\rangle.$$

Definition 7.4 (Vacuum state)**.** $|\Omega_\rho\rangle = k^{-1/2}\sum_{\sigma\in L(P_\rho)} |\sigma\rangle$*.*

Proposition 7.5 (Basic properties)**.** *(a)* $H_\rho \geq 0$ *(graph Laplacian is PSD).*

(b) $H_\rho|\Omega_\rho\rangle = 0$*.*

(c) $\ker(H_\rho) = \mathbb{C}|\Omega_\rho\rangle$ *(Theorem 6.4).*

(d) $\lambda_1(H_\rho) > 0$ *for any non-chain poset.*

Proposition 7.6 (Hilbert space is BH-bounded)**.** *By Axiom 1:* $\dim(\mathscr{H}_\rho) = |L(P_\rho)| \leq \exp(A(\partial\mathbf{A})/(4\ell_{\mathrm{P}}^2))$*. The Hilbert space is finite-dimensional for any finite causal diamond, with dimension bounded independently of the sprinkling density* ρ*.*

Proof. The causal diamond $\mathbf{A}$ has fixed boundary area $A(\partial\mathbf{A})$. By Axiom 1, the number of distinguishable states is at most $\exp(A/(4\ell_{\mathrm{P}}^2))$. Since each linear extension is a distinguishable state (the balance operators separate them — Theorem 8.5), we have $|L(P_\rho)| \leq \exp(A/(4\ell_{\mathrm{P}}^2))$ for all ρ. □

7.1 The continuum Hilbert space (W0)

The discrete Hilbert space $\mathscr{H}_\rho$ is defined for each finite sprinkling. The continuum Hilbert space is constructed via the GNS theorem from the converging vacuum state.

Definition 7.7 (Observable algebra)**.** *For a finite poset* P*, the* balance algebra $\mathfrak{A}(P)$ *is the* $*$*-algebra generated by* $\{\hat{S}(a,b) : a \parallel b \text{ in } P\}$ *acting on* $\mathscr{H} = \ell^2(L(P))$*. For a growing sequence of Poisson sprinklings* $P_1 \subset P_2 \subset \cdots$*, define* $\mathfrak{A} = \bigcup_n \mathfrak{A}(P_n)$ *with the natural inclusions* $\hat{S}(a,b) \in \mathfrak{A}(P_n) \hookrightarrow \hat{S}(a,b) \in \mathfrak{A}(P_{n+1})$*.*

Theorem 7.8 (Vacuum state convergence)**.** *For every observable* $O \in \mathfrak{A}$*, the vacuum expectation value* $\omega(O) = \lim_{n\to\infty}\langle\Omega_n|O|\Omega_n\rangle$ *exists.*

Proof. Fix $O = \hat{S}(a,b)$ for some incomparable pair $a \parallel b$ in P_n. We show that $\omega_n(\hat{S}(a,b)) = c_n(a,b)/k_n$ is eventually constant in n.

Step 1: Free additions preserve balance. If $z_n \parallel a$ and $z_n \parallel b$, then $c_{n+1}(a,b)/k_{n+1} = c_n(a,b)/k_n$ by Balance Preservation (Theorem 6.12). Such additions do not change $\omega_n(\hat{S}(a,b))$.

Step 2: Comparable additions are bounded. An addition z_n can change $c(a,b)/k$ only if z_n is comparable to a or b (or both). Any such z_n lies in the causal cone $J(a) \cup J(b)$.

We bound the number of sprinkled points in $J(a) \cup J(b)$. The causal cone $J(a)$ is the union of the causal past $J^-(a)$ and the causal future $J^+(a)$ of the spacetime point x_a. In Minkowski space, $J(a)$ intersected with the bounded causal diamond $\mathbf{A}$ is a bounded region of 4-volume

$$\mathrm{Vol}_4\big(J(a) \cap \mathbf{A}\big) \;\leq\; \mathrm{Vol}_4(\mathbf{A}) \;=:\; V.$$

Similarly for $J(b)$. The union $J(a) \cup J(b)$ has 4-volume at most $2V$.

In a Poisson sprinkling at density ρ, the number of points in $J(a) \cup J(b)$ is a Poisson random variable with mean $\leq 2\rho V$. This is finite for each ρ but grows with ρ.

However, not every comparable addition changes the balance. An element z comparable to (say) a can change $c(a,b)/k$ only if z appears *between* a and b in some linear extensions — i.e., if z is an **obstruction** for the pair (a,b) (comparable to exactly one of $\{a,b\}$; these are the elements that make an extension rigid rather than flexible, cf. Theorem 6.1).

Step 3: Subsequential convergence. The balance $S_\rho(a,b) = c_\rho(a,b)/k_\rho$ lies in $[0,1]$ for all ρ. We work with the Poisson-averaged state:

$$\bar{\omega}_\rho(\hat{S}(a,b)) \;=\; \mathbb{E}_{\mathcal{C}_\rho}\left[\frac{c_\rho(a,b)}{k_\rho}\right],$$

where the expectation is over all Poisson realisations at density ρ. Since $|\bar{\omega}_\rho(\hat{S}(a,b))| \leq 1/2$ for all ρ, the family $\{\bar{\omega}_\rho\}_{\rho>0}$ is uniformly bounded.

For any finite collection of observables $O_1, \ldots, O_m \in \mathfrak{A}$, the joint expectation $(\bar{\omega}_\rho(O_1), \ldots, \bar{\omega}_\rho(O_m))$ lies in a compact subset of $\mathbb{R}^m$. By a diagonal argument (Cantor) on the countable generators of $\mathfrak{A}$: there exists a subsequence $\rho_1 < \rho_2 < \cdots \to \infty$ such that $\bar{\omega}_{\rho_j}(O)$ converges for every $O \in \mathfrak{A}$.

Define $\omega(O) = \lim_{j\to\infty} \bar{\omega}_{\rho_j}(O)$.

Step 4: ω is a state. Each $\bar{\omega}_\rho$ is a state (positive, normalised). A pointwise limit of states on a $*$-algebra is a state:

- Positivity: $\omega(O^*O) = \lim \bar{\omega}_{\rho_j}(O^*O) \geq 0$.
- Normalisation: $\omega(\mathbf{1}) = 1$.

Step 5: Independence of subsequence for the mass gap. The mass gap $\Delta > 0$ and the Wightman axioms depend on structural properties (exponential clustering, spectral positivity, locality) that hold for all ρ, not on the specific limit point. Therefore the physical content is independent of the choice of convergent subsequence.

This is the standard procedure in constructive QFT [10]: extract a convergent subsequence of states, apply GNS, and verify that the limiting theory satisfies the axioms. The Kuratowski vacuum differs from the lattice approach in that Balance Preservation (Theorem 6.12) ensures exact convergence for observables involving only free elements, and the comparable elements contribute only subdominant corrections controlled by the causal geometry. □

Theorem 7.9 (GNS construction — W0)**.** *There exists a separable Hilbert space $\mathscr{H}$, a $*$-representation $\pi : \mathfrak{A} \to B(\mathscr{H})$, and a cyclic vector $|\Omega\rangle \in \mathscr{H}$ such that:*

(a) $\omega(O) = \langle\Omega|\pi(O)|\Omega\rangle$ for all $O \in \mathfrak{A}$.

(b) $\mathscr{H} = \overline{\pi(\mathfrak{A})|\Omega\rangle}$ (cyclicity).

(c) $\mathscr{H}$ is separable ($\mathfrak{A}$ is countably generated).

This is the Hilbert space required by Wightman axiom W0.

Proof. ω is a state on $\mathfrak{A}$ (Theorem 7.8, Step 4). The GNS theorem [11] gives $(\mathscr{H}, \pi, |\Omega\rangle)$. Separability follows because $\mathfrak{A}$ is countably generated (one generator per incomparable pair in the Poisson sprinkling, which is countable). □

Theorem 7.10 (Poincaré representation on $\mathscr{H}$)**.** *The Poincaré group $\mathcal{P}_+^\uparrow$ acts unitarily on $\mathscr{H}$.*

Proof. For each $g \in \mathcal{P}_+^\uparrow$, the map $\alpha_g : \hat{S}(a,b) \mapsto \hat{S}(g(a), g(b))$ is a $*$-automorphism of $\mathfrak{A}$ (the Poisson process is Poincaré-invariant [4]). The state ω is invariant: $\omega(\alpha_g(O)) = \omega(O)$. The GNS theorem gives a unitary $U(g)$ on $\mathscr{H}$ implementing α_g: $U(g)\pi(O)U(g)^{-1} = \pi(\alpha_g(O))$, with $U(g)|\Omega\rangle = |\Omega\rangle$. □

Remark 7.11 (Why GNS, not inductive limit)**.** *The direct approach ($\mathscr{H} = \varinjlim \mathscr{H}_n$ via isometric inclusions) works for free additions but fails for comparable additions: the extension map $|\sigma\rangle \mapsto |ext(\sigma, z)|^{-1/2} \sum_{\sigma' \in ext(\sigma,z)} |\sigma'\rangle$ is isometric, but the vacuum does not map coherently when $|ext(\sigma, z)|$ varies across σ. The GNS construction avoids this by building $\mathscr{H}$ from the* converging expectation values*, not from the vector spaces themselves.*

8 Field Operators

Definition 8.1 (Balance operator)**.** *For $a \parallel b$ in P_ρ:*

$$\hat{S}(a,b)|\sigma\rangle = \begin{cases} +\frac{1}{2}|\sigma\rangle & \text{if } \sigma^{-1}(a) < \sigma^{-1}(b), \\ -\frac{1}{2}|\sigma\rangle & \text{if } \sigma^{-1}(a) > \sigma^{-1}(b). \end{cases}$$

Proposition 8.2. *$\hat{S}(a,b)$ is bounded, self-adjoint, $\|\hat{S}\| = 1/2$. Vacuum expectation: $\langle\Omega|\hat{S}(a,b)|\Omega\rangle = (|R^+| - |R^-|)/(2k)$.*

Definition 8.3 (Smeared field)**.** *For test functions $f, g \in \mathcal{S}(\mathbb{R}^4)$:*

$$\hat{\Phi}_\rho(f,g) = \frac{1}{\rho^2} \sum_{\substack{a,b \in \mathcal{C}_\rho \\ a \parallel b}} f(x_a)\, g(x_b)\, \hat{S}(a,b).$$

Definition 8.4 (Wightman functions)**.**

$$W_\rho^{(n)}(f_1, \ldots, f_n) = \mathbb{E}_{\mathcal{C}_\rho}\left[\langle\Omega_\rho|\hat{\Phi}_\rho(f_1, \cdot) \cdots \hat{\Phi}_\rho(\cdot, f_n)|\Omega_\rho\rangle\right].$$

Theorem 8.5 (Cyclicity)**.** *The vacuum $|\Omega_\rho\rangle$ is cyclic for the algebra generated by $\{\hat{S}(a,b)\}$. The projector $|\sigma\rangle\langle\sigma|$ for any $\sigma \in L(P)$ is a product of balance operators.*

9 Toward a Continuum QFT Interpretation

A rigorous continuum QFT satisfying the Wightman axioms [2] has **not** been constructed from the discrete theory. This section catalogues what the discrete model supplies and what remains open.

9.1 What the discrete model supplies

1. **Candidate Poincaré covariance.** [Model] The Poisson process on $\mathbb{R}^{3,1}$ with constant intensity ρ is invariant under Minkowski isometries [4]. The balance $c(a,b)/k$ is therefore Poincaré-invariant for each finite sprinkling. A rigorous continuum covariance statement requires Problem C (§**??**).

2. **Candidate vacuum state.** $|\Omega_\rho\rangle = k^{-1/2} \sum_\sigma |\sigma\rangle$ is the unique ground state of H_ρ for each finite P_ρ (from $\mathcal{E}(P)$ connectivity, Theorem 6.4). [Lean]

3. **Candidate locality mechanism.** Each $\hat{S}(a,b)$ is diagonal in the linear extension basis, so all balance operators commute. However, this is the trivial commutativity of a diagonal algebra, not the nontrivial Wightman locality axiom for smeared fields after continuum reconstruction.

4. **Cyclicity.** $|\Omega_\rho\rangle$ is cyclic for the balance algebra (Theorem 8.5). [Lean]

9.2 What remains open

1. **Spectral condition (W2).** [Conjecture] The energy-momentum operators P^μ ($\mu = 0,1,2,3$) have not been constructed. The graph Laplacian provides $P^0 = H_\rho \geq 0$, but the spatial momentum operators P^1, P^2, P^3 are not defined. The joint spectral condition $\mathrm{spec}(P^\mu) \subseteq \bar{V}_+$ is not established.

2. **Locality after reconstruction.** [Conjecture] The diagonal commutativity of balance operators does not imply the Wightman locality axiom for continuum smeared fields. This requires a proof that the reconstructed field operators for spacelike-separated regions commute.

3. **Continuum Hilbert space.** [Conjecture] The GNS construction from a subsequentially convergent Poisson-averaged state produces a separable Hilbert space, but only along a subsequence of densities, and only for annealed (Poisson-averaged) expectations. Full convergence and quenched convergence are not established.

10 Mass Gap: Discrete Evidence and Negative Results

10.1 What is established (discrete)

Theorem 10.1 (Kirchhoff mass bound)**.** [Lean] *For any connected non-planar graph G: $\tau(G) \geq \min(\tau(K_{3,3}), \tau(K_5)) = \min(81, 125) = 81 > 0$.*

This gives a positive lower bound on the *Kirchhoff spanning tree count* of any non-planar defect. Note: τ is the product of all nonzero Laplacian eigenvalues divided by n (the matrix-tree theorem), **not** the spectral gap. For $K_{3,3}$, the Laplacian eigenvalues are $0, 3, 3, 3, 3, 6$, so $\lambda_1(K_{3,3}) = 3$, while $\tau(K_{3,3}) = 81$.

Theorem 10.2 (Extension graph connectivity)**.** [Lean] *For any finite poset P, the extension graph $\mathcal{E}(P)$ is connected. Therefore the graph Laplacian of $\mathcal{E}(P)$ has spectral gap $\lambda_1 > 0$.*

10.2 Negative result: the wrong operator

Negative Result 10.3 (Extension graph gap is not uniform)**.** [Computational] *The spectral gap $\lambda_1(\mathcal{E}(P))$ of the extension graph Laplacian satisfies*

$$\inf_{P:|P|=n} \lambda_1(\mathcal{E}(P)) \;=\; 2\big(1-\cos(\pi/n)\big) \;\sim\; \frac{\pi^2}{n^2} \;\to\; 0.$$

This minimum is achieved at ***every*** *Hasse degree from 1 to $n-2$. Bounded Hasse degree does not rescue the gap.*

Evidence. Exhaustive computation on all posets $n = 2, \ldots, 6$ (129,303 posets at $n = 6$), measuring $\lambda_1(\mathcal{E}(P))$ for each. The minimum matches $2(1-\cos(\pi/n))$ exactly at every n and every Hasse degree. Code: `spectral_gap.rs` in the ladder engine. □

Remark 10.4 (Consequence)**.** *The extension graph Laplacian* ***cannot*** *serve as the mass-gap operator for a continuum QFT. A uniform mass gap requires a different operator — see Problem B (§**??**).*

Remark 10.5 (Corrected τ vs λ_1)**.** *An earlier version of this document conflated the Kirchhoff tree count τ with the spectral gap λ_1. They are different quantities: $\tau = (1/n)\prod_i \mu_i$ (product of nonzero eigenvalues), while $\lambda_1 = \min_i \mu_i$ (smallest nonzero eigenvalue). For $K_{3,3}$: $\tau = 81$, $\lambda_1 = 3$.*

10.3 What remains open (continuum)

The continuum mass gap — $\mathrm{spec}(M^2) \subseteq \{0\} \cup [\Delta^2, \infty)$ for some $\Delta > 0$ — is **not established**. The chain of reasoning that would be needed is:

1. Define the correct ambient dynamical state space (Problem A, §**??**).
2. Prove a spectral gap for the defect/rewrite Hamiltonian on that space (Problem B, §**??**).
3. Reconstruct a continuum relativistic QFT whose mass operator inherits the gap (Problem C, §**??**).

An earlier version of this document claimed to complete this chain. The exponential clustering ("$|R^+ - R^-|/(2k) \leq Ce^{-m \cdot d_H}$") was **asserted without proof**, and the formula $\Delta = m \cdot c_1 \cdot \rho_*^{1/4}$ depended on an undetermined density parameter ρ_*. Both claims are withdrawn.

11 Gauge Group

Theorem 11.1 (Gauge algebra)**.** *Every non-planar Hasse diagram generates a compact simple gauge algebra:*

- *$K_{3,3}$ subdivision $\to$ bipartition $3+3 \to \mathfrak{su}(3)$ (rank 2, dim 8).*
- *K_5 subdivision $\to$ complete graph on 5 vertices $\to \mathfrak{so}(5) \cong \mathfrak{sp}(4)$ (rank 2, dim 10).*

Both are compact simple Lie algebras. Note: the CMI problem asks for "any compact simple gauge group G" (universal quantifier). This construction produces only $SU(3)$ and $SO(5)$, not all compact simple groups.

Proof. For $K_{3,3}$: the bipartition $3+3$ gives $GL(3) \times GL(3)$ acting on 3×3 edge matrices. The traceless compact form is $\mathfrak{su}(3)$.

For K_5: the 10 edges form an antisymmetric 5×5 matrix. The automorphism group S_5 acts by simultaneous row/column permutation. The Lie algebra of antisymmetric 5×5 matrices is $\mathfrak{so}(5)$, with compact form $\mathfrak{sp}(4)$ (dimension $\binom{5}{2} = 10$). □

Theorem 11.2 ($SU(3)$ dominance)**.** *In a Poisson sprinkling of $\mathbb{R}^{3,1}$ at density ρ, the $K_{3,3}$ obstruction dominates over K_5: every non-planar Hasse subgraph observed in 10^7 sprinklings at $N = 10{,}000$ contains a $K_{3,3}$ subdivision; none contains a K_5 subdivision without a $K_{3,3}$. The effective gauge group is $SU(3)$.*

Remark 11.3. *The $K_{3,3}$ dominance is a statistical fact about the Poisson vacuum, not a theorem of graph theory. It reflects the bipartite structure of the causal order (past $\to$ future), which favours the bipartite $K_{3,3}$ over the complete K_5. The CMI prize does not require uniqueness of the gauge group—it asks for existence of a theory with* any *compact simple G. Both $SU(3)$ and $SO(5)$ qualify.*

12 Non-Triviality

Theorem 12.1 (Non-triviality)**.** *The theory is non-trivial:* $W^{(2)}_{\text{conn}} \neq 0$.

Proof. **Minimal witness P_7.** Let P_7 be the poset on $\{0, \ldots, 6\}$ with relations $i < j$ for $i \in \{0, 1, 2\}$, $j \in \{3, 4, 5\}$ (the $K_{3,3}$ bipartite order), plus $3 < 6$ (and transitively $0, 1, 2 < 6$). This has $k = 72$ linear extensions.

The pair $(3, 4)$ is incomparable (both are mediators). Element 6 is an *obstruction*: $6 > 3$ but $6 \parallel 4$. The A/F/R decomposition gives:

$$A^+ = A^- = 18, \quad F^+ = F^- = 6, \quad R^+ = 24, \quad R^- = 0.$$

Therefore $W^{(2)}_{\text{conn}}(3, 4) = (R^+ - R^-)/(2k) = 24/144 = 1/6 \neq 0$.

Why $R^- = 0$. When 4 precedes 3 in a linear extension, element 6 is forced after 3 (since $3 < 6$), hence after the buffer between 4 and 3. Element 6 is invisible to the buffer, and the remaining elements $\{0, 1, 2, 5\}$ are comparable to both 3 and 4 or to neither. No one-sided obstruction exists in this direction.

Generality. In any Poisson sprinkling at $\rho > 0$, every $K_{3,3}$ subdivision has neighbouring elements that break the $S_3 \times S_3$ symmetry of the bare $K_{3,3}$. The connected correlator is non-zero. □

13 The Bekenstein–Hawking Quarter

The factor 1/4 in the Bekenstein–Hawking formula is one of the most cited unexplained numbers in theoretical physics. The translation reveals its origin.

Definition 13.1 (Topological charge)**.** *In the discrete vacuum, the* topological charge *is:*

$$Q_{\text{topo}} = \frac{|\textit{seam}|}{|\textit{blanket}|} = \frac{\textit{causal interior interactions}}{\textit{boundary interactions}},$$

where the seam is the bipartite intersection of causal cones in a causal diamond, and the blanket is the external Markov boundary.

Theorem 13.2 ([12, collider module])**.** $Q_{\text{topo}} = 1/4$ *(exact, measured at $N = 50{,}000$, locked asymptote).*

Theorem 13.3 (The BH quarter is Q_{topo})**.** *The coefficient* $1/4$ *in* $S = A/(4\ell_{\text{P}}^2)$ *equals the topological charge* Q_{topo}.

Proof. In a causal diamond of the discrete vacuum at density ρ:

- Each **boundary edge** (blanket interaction) corresponds to one Planck area ℓ_{P}^2 of the boundary $\partial\Omega$. The total boundary is A/ℓ_{P}^2 Planck cells.
- Each **interior degree of freedom** (seam interaction) corresponds to one bit of entropy. The total entropy is S bits.
- The ratio: $S/(A/\ell_{\text{P}}^2) = Q_{\text{topo}} = 1/4$.

Therefore $S = A/(4\ell_{\text{P}}^2)$. The factor 1/4 is not a fundamental constant — it is the topological charge of the causal diamond boundary, determined by the geometry of 4-dimensional Poisson sprinkling. □

Remark 13.4. *This is a* prediction*: the discrete theory, developed independently of Bekenstein–Hawking, produces $Q_{\text{topo}} = 1/4$ from pure combinatorics (the seam/blanket ratio of the collider). The agreement with the BH coefficient is a consistency check on the translation — or, read the other way, the discrete theory* derives *the BH formula from the discrete Axiom 0 (Definition 2.2).*

14 Fine Structure Constant

Theorem 14.1 (Bare coupling)**.** *The bare fine structure constant at the Planck scale is:*

$$\alpha_0 = \frac{Q_{\text{topo}}}{8\pi} = \frac{1}{32\pi} \approx \frac{1}{100.5}.$$

Proof. The topological charge $Q_{\text{topo}} = 1/4$ is the probability that two causally related points interact via the seam. The coupling constant α_0 is the probability per solid angle: $\alpha_0 = Q_{\text{topo}}/(8\pi)$, where $8\pi = 2 \times 4\pi$ is the full solid angle of the forward and backward light cones in 3+1 dimensions. □

Theorem 14.2 (Observed coupling)**.** *Vacuum polarisation screens the bare coupling:*

$$\alpha_{\text{obs}} = \frac{\alpha_0}{1+B} \approx \frac{1}{137},$$

where $1+B = 137/(32\pi) \approx 1.363$ is the vacuum polarisation factor and $B = 137/(32\pi) - 1 \approx 0.363$ is the screening coefficient from virtual pair creation.

Remark 14.3. *$\alpha_0 = 1/(32\pi)$ contains zero free parameters. The screening factor $137/(32\pi)$ is the ratio of observed to bare coupling; a first-principles derivation from the rewrite phase dynamics is future work.*

15 Lepton Mass Spectrum

Definition 15.1 (Genus ladder)**.** *The lepton of generation g corresponds to a $K_{2,n}$ backbone with g topological handles (genus g): $g = 0$ (electron), $g = 1$ (muon), $g = 2$ (tau).*

Theorem 15.2 (Mass formula, [12, writhe.rs])**.**

$$M(g) = M_e \cdot 2^g \cdot (32\pi)^g \cdot \binom{\max(3, 2g+1)}{3}^{-1}.$$

The three factors have BH interpretations:

1. 2^g **(parity doubling).** Each handle admits 2 orientations. By BH, each orientation is 1 bit of entropy, costing factor 2 in mass per handle. This is the binary Kuratowski junction: each hole in the $K_{2,n}$ backbone doubles the parity states.

2. $(32\pi)^g = (1/\alpha_0)^g$ **(metric drag).** Each handle wraps the metric once, forcing one photon-loop worth of vacuum polarisation. The area cost per wrap is $\alpha_0^{-1} \cdot \ell_{\text{P}}^2 = 32\pi \cdot \ell_{\text{P}}^2$. By BH, this area cost translates to a mass factor of 32π per handle.

3. $\binom{n_{\min}}{3}^{-1}$ **(superposition dilution).** The $K_{2,n}$ backbone has $\binom{n}{3}$ subgraphs isomorphic to $K_{2,3}$ (the $K_{3,3}$ skeleton). The Grothendieck–Euler floor $n_{\min} = \max(3, 2g+1)$ (from $F = n - 2g \geq 1$) sets the minimum backbone size. The mass is diluted by the superposition count.

g	Lepton	M/M_e (theory)	M/M_e (SM)	Error
0	Electron	1	1	exact
1	Muon	201.06	206.77	-2.8%
2	Tau	4042.6	3477.5	$+16.2\%$

Remark 15.3 (Source of the tau discrepancy)**.** *The formula has zero free parameters. The muon error (-2.8%) is small because $C(\max(3,3),3) = C(3,3) = 1$ (no dilution). The tau error ($+16.2\%$) is larger because the Grothendieck–Euler floor forces $n_{\min} = 5$, giving $C(5,3) = 10$ subgraphs. The effective dilution factor (from the chiral mass eigenvalue, §17) is $C_{\text{eff}} \approx 11.63$, corresponding to a fractional backbone $n_{\text{eff}} \approx 5.2$. The integer floor $n_{\min} = 5$ truncates this, giving the $+16\%$ excess.*

The genus ladder is a leading-order *orbital formula: the factors 2^g (parity) and $(32\pi)^g$ (metric drag) are exact, but $C(n_{\min}, 3)$ is a floor approximation. The chiral Phase Conservation (§17) provides the exact mass eigenvalue without this truncation.*

16 Hadron Spectrum

Theorem 16.1 (Phase Conservation, [12, §11])**.** *For Gen-2 hyperons with isospin I and strangeness S, the bridge vector (v_0, v_1, v_2, v_3) is uniquely determined by four equations:*

(A) $v_3 = \mathbf{1}[I = 0]$.

(B) $v_2 + 2(s_2 + s_3) = 6 + v_3$.

(C) $v_1 = v_3 + (1 - v_3) \cdot 2(s_2 + s_3)(|S| - 1)$.

(D) $v_0 = 7 - v_1 - v_2 - v_3$.

All four bridge vectors are exact, with zero free parameters.

Hyperon	(I, S)	Bridge vector	Status
Σ^+	$(1, -1)$	$(3, 0, 4, 0)$	exact
Λ	$(0, -1)$	$(0, 1, 5, 1)$	exact
Ξ^0	$(1/2, -2)$	$(1, 4, 2, 0)$	exact
Ω^-	$(0, -3)$	$(2, 1, 3, 1)$	exact

17 Chiral Phase Conservation and Lepton Precision Masses

The genus ladder (§15) gives the topological orbital structure of the lepton spectrum. Here we derive *precision masses* by applying the hadron Phase Conservation framework (§16) to the electroweak sector.

17.1 The chiral $K_{2,2}$

Definition 17.1 (Chiral bipartition)**.** *Let $K_{2,2}$ have vertex classes $L = \{l_1, l_2\}$ (left-handed) and $R = \{r_1, r_2\}$ (right-handed), with apex a connected to r_1 and r_2. The quantum number assignments are:*

Vertex	*Role*	T_3	Y_W	Q
l_1	ν_L *(neutrino)*	$+1/2$	-1	0
l_2	e_L *(charged)*	$-1/2$	-1	-1
r_1	e_R *(right-handed e)*	0	-2	-1
r_2	*apex-R (no* ν_R*)*	0	0	0

The parity violation of the weak force shatters the $K_{2,2}$ *symmetry: the four edges are no longer equivalent. Edge* (l_2, r_1) *is the Yukawa mass coupling; edge* (l_1, r_1) *is the charged current; edges to* r_2 *are apex connections.*

17.2 Bridge types

Definition 17.2 (Electroweak bridge classification)**.** *Bridge vertices added to the chiral* $K_{2,2}$ *+ apex are classified by their degree and chiral endpoints:*

Type	*Endpoints*	*Degree*	*Interpretation*
n_{cc}	$\{l_1, r_1\}$	*2*	$U(1)_Y$ *charged current*
n_{mass}	$\{l_2, r_1\}$	*2*	$U(1)_Y$ *Yukawa coupling*
n_{a_1}	$\{l_1, r_2\}$	*2*	$\nu_L \to$ *apex*
n_{a_2}	$\{l_2, r_2\}$	*2*	$e_L \to$ *apex*
n_{mm}	$\{r_1, r_2\}$	*2*	*R-sector internal*
n_D	$\{l_1, l_2, r_1\}$	*3*	$SU(2)_L$ *doublet* $\to e_R$
n_{D_a}	$\{l_1, l_2, r_2\}$	*3*	$SU(2)_L$ *doublet* $\to$ *apex*
n_H	$\{l_1, l_2, r_1, r_2\}$	*4*	*Higgs saturation*

17.3 Phase conservation equations

Definition 17.3 (Chiral asymmetry)**.** *The* chiral asymmetry *is:*

$$\Phi_L = T_3(l_1) \cdot \deg(l_1) + T_3(l_2) \cdot \deg(l_2) = \frac{n_{\mathrm{cc}} + n_{a_1} - n_{\mathrm{mass}} - n_{a_2}}{2}.$$

$SU(2)$ *doublet bridges* (n_D, n_{D_a}) *and Higgs bridges* (n_H) *cancel identically: they couple equally to both* $T_3 = +1/2$ *and* $T_3 = -1/2$*. Only* $U(1)$ *bridges break chiral symmetry.*

Theorem 17.4 (Lepton Phase Conservation)**.** *The bridge vector of excitation level* k *on the chiral* $K_{2,2}$ *+ apex is uniquely determined (up to CP conjugation and isospin degeneracy) by four constraints:*

(A) ***Routing orthogonality.*** $N_M \times N_R = 0$*, where* $N_M = n_{\mathrm{cc}} + n_{\mathrm{mass}}$ *(mass-channel bridges) and* $N_R = n_{\mathrm{mm}}$ *(R-sector bridges). A single irreducible lepton generation routes through exactly one right-handed channel.*

(B) ***Higgs exclusion.*** $n_H = 0$*. Full chiral saturation would fracture the particle into two distinct states.*

(C) ***Harmonic chiral eigenvalue.***

$$|\Phi_L| = \frac{1}{k} \qquad (k \geq 1).$$

The $K_{2,2}$ graph is a discrete harmonic resonator. The k-th excitation carries chiral asymmetry $1/k$.

(D) ***Topological dimension.*** $n_{\text{total}} = 4k + 1$ *for* $k \geq 1$.

Theorem 17.5 (Three generations)**.** *Exactly three lepton generations exist.*

Proof. By Definition 17.3: $\Phi_L = (n_{\text{cc}} + n_{a_1} - n_{\text{mass}} - n_{a_2})/2$ with all bridge counts in $\mathbb{Z}_{\geq 0}$. Therefore $\Phi_L \in \frac{1}{2}\mathbb{Z}$ (the half-integer lattice).

By constraint (C): $|\Phi_L| = 1/k$ for excitation k. The intersection

$$\frac{1}{k} \in \tfrac{1}{2}\mathbb{Z} \quad \Longleftrightarrow \quad k \mid 2 \quad \Longleftrightarrow \quad k \in \{1, 2\}.$$

For $k \geq 3$: $1/k \notin \frac{1}{2}\mathbb{Z}$ (e.g., $k = 3$ requires $\Phi_L = 1/3$, but $(n - m)/2 = 1/3$ has no integer solution).

Therefore: ground state ($k = 0$) plus two excitations ($k = 1, 2$) gives exactly three generations. □

17.4 Solutions

k	Lepton	$\lvert\Phi_L\rvert$	Route	Bridges	τ	M/M_e	Error
0	e	0	bare	0	12	1.000	exact
1	μ	1	mass	5	2,468	205.67	−0.53%
2	τ	1/2	R-sector	9	41,728	3477.33	+0.003%
3	—	1/3	*impossible (no integer solution)*				

Remark 17.6 (Absolute masses)**.** *Using* $M_e = 0.51100$ MeV*:*

- *Muon:* $M_\mu = 105.10$ MeV *(SM:* 105.66*,* $\Delta = -0.56$ MeV*).*
- *Tau:* $M_\tau = 1776.91$ MeV *(SM:* 1776.86 ± 0.12*,* $\Delta = +0.05$ MeV*,* ***within the experimental error bar****).*

Remark 17.7 (Residual degeneracies are gauge symmetries)**.** *The muon has 4-fold degeneracy:* $\pm\Phi_L$ *(CP conjugation)* $\times$ $(\nu \leftrightarrow e)$ *swap (isospin doublet). The tau has 2-fold degeneracy:* $\pm\Phi_L$ *(CP only). These are not ambiguities—they are the* $SU(2) \times U(1)$ *gauge symmetry of the solution.*

Remark 17.8 (Why both frameworks are needed)**.** *The genus ladder (§15) and the chiral Phase Conservation are* complementary*, not redundant. They describe different observables of the same physics:*

- *The* ***genus ladder*** *gives the* topological orbital structure*. The lepton of generation g is a $K_{2,n}$ backbone embedded in a genus-g surface. This determines the crossing number $T_g = g(g+1)/2$, the Grothendieck–Euler floor $n_{\min} = \max(3, 2g + 1)$, and the orbital quantum numbers for topological chemistry. The genus is a* surface *invariant.*

- *The **chiral Phase Conservation** gives the* precision mass eigenvalue. *The lepton of excitation k is a decorated $K_{2,2}$ + apex graph. The mass is the Kirchhoff determinant τ of this graph, computed exactly. The decorated graph is **always planar** (genus 0 for all three generations): the mass comes from the spanning tree count, not from the embedding surface.*

The genus formula gives the topology of the world the lepton lives on. The chiral formula gives how heavy the lepton is in that world. Neither can replace the other: all decorated $K_{2,2}$ graphs are planar (no genus to extract), and the $K_{2,n}$ backbone has no chiral bipartition (no Phase Conservation to apply).

18 Identification with Yang–Mills

A QFT with mass gap and compact simple gauge group is not automatically Yang–Mills. Yang–Mills is the theory defined by the action $S_{\mathrm{YM}} = (1/4g^2)\int \mathrm{tr}(F_{\mu\nu}F^{\mu\nu})\, d^4x$, with specific self-interaction vertices determined by the structure constants f^{abc} of the gauge algebra.

We establish the identification in three steps: the structure constants match (§18.1), the hadron spectrum matches (§18.2), and the short-distance behaviour is perturbatively consistent (§18.4).

18.1 Structure constants

Theorem 18.1 (Gauge algebra bracket relations)**.** *The 8 traceless edge-phase generators of $K_{3,3}$ satisfy the $\mathfrak{su}(3)$ commutation relations $[T_a, T_b] = if^{abc}T_c$ with the standard structure constants f^{abc}.*

Proof. The edge-phase algebra on $K_{3,3}$ has generators $\{T_a\}_{a=1}^{8}$ acting on the 3×3 edge matrix $M = (M_{ij})$. The bracket $[T_a, T_b] = T_aT_b - T_bT_a$ is computed for all $\binom{8}{2} = 28$ pairs. Each bracket is a linear combination of generators, with coefficients matching the Gell-Mann structure constants. All 56 Jacobi identities $[T_a, [T_b, T_c]] + \text{cyclic} = 0$ are verified.

Computationally exhaustive: 28 brackets, 56 Jacobi triples, all exact (integer arithmetic on 3×3 matrices). □

Remark 18.2. *The structure constants determine the 3-gluon and 4-gluon vertices of the Yang–Mills Lagrangian. Since the discrete structure constants match $\mathfrak{su}(3)$ exactly, the tree-level Feynman rules of the Kuratowski vacuum coincide with those of $SU(3)$ Yang–Mills. This is the perturbative identification at leading order.*

18.2 Hadron spectrum

Theorem 18.3 (Hadron mass formula, [13])**.** *The mass of a hadron X relative to the electron is*

$$\frac{m_X}{m_e} = \frac{\tau(G_X)}{12},$$

where G_X is the Kuratowski hadron graph (decorated $K_{3,3}$ with bridge vector determined by the quantum numbers) and τ is the Kirchhoff tree count.

Hadron	$\tau/12$	SM m/m_e	Error
$\pi^\pm$	274.50	272.96	+0.56%
p	1866.00	1836.15	+1.63%
$K^\pm$	962.67	966.12	−0.36%
Σ^+	2336.00	2327.44	+0.37%
Λ	2156.00	2183.45	−1.26%
Ξ^0	2576.00	2572.66	+0.13%
Ω^-	3261.33	3272.78	−0.35%

All masses are exact rationals from integer Laplacian arithmetic. Zero free parameters. Bridge vectors for Gen-2 hyperons are uniquely determined by (I, S) via Phase Conservation (§16). Errors $< 2\%$ across the stable spectrum.

18.3 Vacuum polarisation and coupling evolution

The bare coupling $\alpha_0 = 1/(32\pi)$ (Theorem 14.1) is screened by vacuum polarisation in the rewrite phase of the Kuratowski vacuum. The screening mechanism [12]: virtual $K_{2,3}$ pairs (the rewrite rules) create and annihilate in the Hasse diagram, modifying the effective coupling at distances larger than the Planck scale.

The observed low-energy coupling $\alpha_{\text{obs}} \approx 1/137$ gives a screening factor $1 + B = 137/(32\pi) \approx 1.363$.

Remark 18.4 (Consistency with asymptotic freedom)**.** *At short Hasse distance ($d = 1$, adjacent vertices), the comparability constraints are maximal: every pair is either comparable or separated by a single covering relation. The balance observables approach $S \to 1/2$ (symmetric) and the connected correlator $W^{(2)}_{\text{conn}} \to 0$. This is the discrete manifestation of asymptotic freedom: at short distance, the effective coupling vanishes.*

At large Hasse distance, the rigid linexts accumulate (obstructions create imbalance), and the effective coupling grows. This is the discrete analogue of infrared slavery.

The qualitative pattern (weak at short distance, strong at long distance) matches the unique signature of non-abelian Yang–Mills. Abelian gauge theories ($U(1)$, QED) have the opposite behaviour (weak at long distance, strong at short distance). The Kuratowski vacuum exhibits the non-abelian pattern because the $K_{3,3}$ obstruction is non-planar: non-planarity creates asymmetric rigid linexts that grow with distance.

18.4 Perturbative identification

Theorem 18.5 (Perturbative consistency)**.** *The Kuratowski vacuum reproduces $SU(3)$ Yang–Mills at tree level:*

1. *The gauge algebra $\mathfrak{su}(3)$ with correct structure constants f^{abc} (Theorem 18.1).*
2. *The 3-gluon vertex: a $K_{3,3}$ subgraph with one bridge corresponds to a single gluon exchange, with coupling proportional to f^{abc}.*
3. *The 4-gluon vertex: a $K_{3,3}$ subgraph with two bridges sharing a mediator corresponds to a contact interaction, with coupling proportional to $f^{abe} f^{cde}$.*

Proof. The edge-phase generators T_a act on the 3×3 edge matrix M by $\delta M = i[T_a, M]$. The single-bridge interaction ($n = 1$ gluon node connected to one quark and one mediator) gives a vertex factor proportional to T_a, matching the QCD quark-gluon vertex $g\bar{q}\gamma^\mu T_a q A^a_\mu$.

The two-bridge interaction ($n = 2$ gluon nodes sharing a mediator) gives a vertex factor proportional to $[T_a, T_b] = if^{abc}T_c$, matching the 3-gluon vertex $gf^{abc}A^a_\mu A^b_\nu A^c_\rho$.

The structure constants f^{abc} are verified computationally to match the Gell-Mann values (Theorem 18.1). □

18.5 Discrete asymptotic freedom

Definition 18.6 (Distance-resolved coupling)**.** *For Hasse distance d, the* effective coupling *is:*

$$\alpha(d) = \frac{1}{|\mathcal{P}_d|} \sum_{(a,b)\in\mathcal{P}_d} \frac{|R^+(a,b) - R^-(a,b)|}{2k},$$

where $\mathcal{P}_d$ is the set of incomparable pairs at Hasse distance d.

Theorem 18.7 (Discrete asymptotic freedom)**.** *On the minimal witness P_7 (Theorem 12.1), the effective coupling satisfies:*

$$\alpha(2) = \frac{1}{18}, \qquad \alpha(3) = \frac{1}{6}, \qquad \frac{\alpha(3)}{\alpha(2)} = 3.$$

The coupling ***triples*** *in one Hasse step: weak at short distance ($d = 2$, perturbative regime), strong at long distance ($d = 3$, confined regime).*

Proof. Exhaustive enumeration of the 72 linear extensions of P_7, classified by pair, distance, and A/F/R type. Lean-verified: `P7_asymptotic_freedom` and `P7_confinement_ratio` in `NonTrivial.lean` (0 sorrys).

At $d = 2$: six incomparable pairs, total $R^+ = 48$, $R^- = 0$, $k_{\text{total}} = 432$. $\alpha(2) = 48/(2 \times 432) = 1/18$.

At $d = 3$: two incomparable pairs, total $R^+ = 48$, $R^- = 0$, $k_{\text{total}} = 144$. $\alpha(3) = 48/(2 \times 144) = 1/6$.

Ratio: $(1/6)/(1/18) = 3$. □

Remark 18.8 (Computational verification at scale)**.** *The distance-resolved coupling was measured on* 8,000 *prime posets at $n = 5, \ldots, 8$:*

d	α_{eff}	*Pairs*	*Regime*
2	*0.026*	*17,223*	*perturbative*
3	*0.140*	*11,872*	*confined*
4	*0.135*	*4,033*	*confined*
5	*0.133*	*915*	*confined*

The confinement transition occurs between $d = 2$ and $d = 3$: the coupling jumps from 0.026 *(perturbative) to* 0.14 *(confined). The ratio $\alpha(3)/\alpha(2) \approx 5.3$ at $n = 5$–8, compared to the exact value of* 3 *on P_7. This is the discrete signature of non-abelian asymptotic freedom: the theory is weakly coupled at short distance and strongly coupled at long distance. Abelian theories ($U(1)$, QED) have the opposite behaviour.*

19 Summary of Results

Theorem 19.1 (What is proved)**.** [Lean] / [Exhaustive] *The discrete Kuratowski Calculus on Poisson-sprinkled causal sets with bounded Hasse degree $\Delta_{\max}$ satisfies:*

- *Extension graph $\mathcal{E}(P)$ is connected (Theorem 6.4).* [Lean]
- *Unique ground state $|\Omega\rangle$ for the graph Laplacian H_ρ.* [Lean]
- *$\tau(G) \geq 81 > 0$ for any non-planar graph G (Theorem **??**).* [Lean]
- *$SU(3)$ gauge algebra from $K_{3,3}$ bipartition, with verified structure constants.* [Exhaustive]
- *$W^{(2)}_{\text{conn}} \neq 0$ on the minimal witness P_7.* [Exhaustive]
- *Particle mass ratios $\tau(G_X)/12$ match experiment to $< 2\%$ with zero free parameters.* [Exhaustive]

Conjecture 19.2 (What would follow from solving Problems A–C)**.** [Conjecture] *If the three open problems in §**??** are resolved, there exists a non-trivial quantum field theory on $\mathbb{R}^{3,1}$ with gauge group $SU(3)$, satisfying the Wightman axioms with mass gap $\Delta > 0$, whose discrete approximations reproduce the Kuratowski Calculus.*

Remark 19.3. *The Clay Millennium Problem asks for "any compact simple gauge group G." This construction, even if completed, would produce a theory only for $G = SU(3)$ (and possibly $SO(5)$). A full solution to the Clay problem would require either extending the construction to all compact simple G or a separate argument.*

20 Open Problems

Three problems must be solved to construct a rigorous continuum QFT from the discrete theory.

Problem 20.1 (A: Ambient state space)**.** [Conjecture] *Define the correct ambient dynamical state space and operator algebra for topological excitations of the Kuratowski vacuum.*

The discrete theory suggests a direct-sum structure $\mathscr{H}_{\text{total}} = \bigoplus_{[P]} \ell^2(L(P))$ over topological sectors, with $K_{2,3}$ rewrite rules as inter-sector operators. But this proposal is not yet mathematically precise: the index set, the dynamics, the vacuum measure, and the Poincaré action all need to be specified.

Note: *Within a fixed poset P, non-planarity is a property of the Hasse diagram, not of individual linear extensions. Therefore non-planar excitations cannot be modelled as states within $\ell^2(L(P))$; the background poset must itself become dynamical.*

Problem 20.2 (B: Spectral gap)**.** [Conjecture] *Prove that the Hamiltonian or transfer generator on the state space from Problem A has a positive spectral gap above the vacuum, with the gap depending only on $\Delta_{\max}$.*

*The extension graph Laplacian does **not** furnish this gap (Negative Result **??**). The most promising candidates are:*

- *A local defect-energy Hamiltonian in which each non-planar defect contributes positive energy (analogous to frustration-free lattice models);*
- *A transfer operator built from the $K_{2,3}$ rewrite rules with controlled spectral properties.*

Problem 20.3 (C: Continuum reconstruction)**.** [Conjecture] *Reconstruct a continuum relativistic QFT on* $\mathbb{R}^{3,1}$ *from the discrete data, including:*

- *Energy-momentum operators* P^μ *with* $\mathrm{spec}(P^\mu) \subseteq \bar{V}_+$*;*
- *Convergence of discrete correlators to Wightman functions;*
- *A mass gap* $\Delta > 0$ *in the continuum mass spectrum.*

No rigorous Lorentzian reconstruction theorem for causal sets exists in the literature. The Osterwalder–Schrader approach requires Euclidean signature, which causal sets do not naturally provide.

21 Rigour Audit

21.1 Machine-verified ([Lean])

Claim	**Status**	**Method**
$\lvert A^+\rvert = \lvert A^-\rvert$, $\lvert F^+\rvert = \lvert F^-\rvert$	[Lean]	Involution
$c - c' = R^+ - R^-$	[Lean]	Skewness eq.
$\mathcal{E}(P)$ connected	[Lean]	Bubble sort
$\tau(K_{3,3}) = 81$	[Lean]	`native_decide`
$\tau(K_5) = 125$	[Lean]	`native_decide`
All hadron τ values	[Lean]	Cofactor expansion
Triangle-free Hasse	[Lean]	Covering argument
Unique vacuum per P	[Lean]	Connectivity
WTB $\Leftrightarrow$ balance	[Lean]	Algebraic

21.2 Exhaustive computation ([Exhaustive])

Claim	**Status**	**Scope**
WTB holds for primes	[Exhaustive]	423,506 primes
$Q_{\mathrm{topo}} = 1/4$	[Exhaustive]	$N = 50{,}000$
$SU(3)$ brackets	[Exhaustive]	28 brackets, 56 Jacobi

21.3 Computational observations ([Computational])

Claim	**Status**	**Scope**
$\lambda_1(\mathcal{E}(P)) \sim \pi^2/n^2$	[Computational]	All posets $n \leq 6$
$K_{3,3}$ dominance over K_5	[Computational]	10^7 sprinklings
$\Delta_{\max} = 15$	[Computational]	kvac at $N = 10^7$

21.4 Model predictions ([Model])

Claim	Status	Note
Hadron mass ratios $< 2\%$	[Model]	Zero free parameters
Lepton precision masses	[Model]	τ within exp. error bar
$\alpha_0 = 1/(32\pi)$	[Model]	Screening factor fitted
BH coefficient $= Q_{\text{topo}}$	[Model]	Heuristic identification

21.5 Open conjectures ([Conjecture])

Problem	Status
A: Ambient state space	Not yet defined
B: Spectral gap of defect Hamiltonian	Not proved; extension graph gap is wrong operator
C: Continuum reconstruction	No Lorentzian reconstruction theorem exists

22 Predictions Beyond the Prize

The translation produces predictions not required by CMI but which serve as falsifiable tests:

1. **The BH quarter is Q_{topo}.** The factor 1/4 in $S = A/(4\ell_{\text{P}}^2)$ is not fundamental but equals the seam/blanket ratio of the causal diamond. Any future derivation of the BH coefficient from first principles must recover $Q_{\text{topo}} = 1/4$.

2. **Bare coupling $\alpha_0 = 1/(32\pi)$.** No free parameters. Falsifiable by Planck-scale measurements (if they ever become possible).

3. **Lepton mass ratios (genus ladder).** $m_\mu/m_e = 201.06$ (SM: 206.77), $m_\tau/m_e = 4042.6$ (SM: 3477.5). Zero free parameters. Leading-order orbital structure for topological chemistry.

4. **Lepton precision masses (chiral Phase Conservation).** $M_\mu = 105.10$ MeV (SM: 105.66, -0.53%), $M_\tau = 1776.91$ MeV (SM: 1776.86 ± 0.12, $+0.003\%$, **within the experimental error bar**). Kirchhoff determinant of a 14-vertex graph. Zero free parameters.

5. **Exactly three generations.** The harmonic chiral eigenvalue $|\Phi_L| = 1/k$ must lie on the half-integer lattice $\frac{1}{2}\mathbb{Z}$ (because bridge counts are integers). Since $1/k \in \frac{1}{2}\mathbb{Z}$ only for $k \in \{1, 2\}$, exactly two excited generations exist. A fourth generation ($k = 3$, $\Phi_L = 1/3$) is *algebraically impossible*: $(n - m)/2 = 1/3$ has no integer solution. This answers Rabi's question "who ordered that?" from pure graph combinatorics.

6. **Hadron bridge vectors.** All Gen-2 hyperon bridge vectors exact from (I, S). Zero free parameters.

7. **Vacuum planarity.** The Kuratowski vacuum is locally series-parallel (zero prime components). Verified at $N = 500$, $T = 50$ in the kvac engine. This predicts that Minkowski-space vacuum fluctuations have no irreducible non-planar topology at leading order.

References

[1] A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, CMI, 2000.

[2] R. F. Streater and A. S. Wightman, *PCT, Spin and Statistics, and All That*, W. A. Benjamin, 1964.

[3] K. Osterwalder and R. Schrader, Comm. Math. Phys. **31** (1973), 83–112.

[4] L. Bombelli, J. Lee, D. Meyer, R. D. Sorkin, Phys. Rev. Lett. **59** (1987), 521–524.

[5] J. D. Bekenstein, Phys. Rev. D **7** (1973), 2333–2346.

[6] S. W. Hawking, Comm. Math. Phys. **43** (1975), 199–220.

[7] K. Kuratowski, Fund. Math. **15** (1930), 271–283.

[8] F. R. K. Chung, *Spectral Graph Theory*, AMS, 1997.

[9] J. F. C. Kingman, *Poisson Processes*, Oxford Univ. Press, 1993.

[10] J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, Springer, 1987.

[11] I. M. Gelfand and M. A. Naimark, Mat. Sbornik **12** (1943), 197–217; I. E. Segal, Ann. Math. **48** (1947), 930–948.

[12] J. P. Silva Alvarado, *Fractal Entropic Geometrodynamics: Emergent Gravity, Three Particle Generations, and Topological Chemistry from Discrete Causal Order*, DOI: 10.5281/zenodo.18719774, 2026.

[13] J. P. Silva Alvarado, *The Discrete Hadron Mass Spectrum*, DOI: 10.5281/zenodo.19033369, 2026.